

AI CAREERS FOR WOMEN

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Workforce Longevity: Designing Data-Led Retention Strategies

Project Domain

[Excel / Power BI]

A Collaborative Initiative

equipping students with practical skills in AI, Data Analytics, and Business
Intelligence

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3. Problem Statement

Employee attrition is not just an HR concern but a strategic business risk that affects productivity, organizational stability, hiring costs, and long-term growth. Many organizations collect extensive employee data, including demographics, compensation details, job roles, performance ratings, and satisfaction metrics. However, this data is often analyzed in isolation and lacks a structured framework to translate insights into actionable retention strategies.

The absence of a data-driven and predictive retention framework makes it difficult for decision-makers to proactively identify high-risk employee segments, understand key drivers of attrition, and take timely action. Additionally, most existing approaches lack integration with automated systems, resulting in delayed insights and reactive decision-making.

This project addresses this gap by designing a structured, data-driven Employee Retention Framework using the IBM HR Analytics Employee Attrition & Performance dataset obtained from Kaggle. Through systematic data cleaning, KPI development, segmentation analysis, predictive modeling, and interactive dashboard design, the project aims to identify critical attrition drivers and transform raw HR data into strategic decision-support insights.

The final outcome is not merely a dashboard, but an integrated retention decision-support system that combines analytics, prediction, and automation to enable HR leaders and management teams to make proactive, timely, and evidence-based workforce planning decisions.

4. Objectives

- To analyze workforce data to identify key business drivers contributing to employee attrition and organizational instability.
- To design a structured, data-driven employee retention framework that enables proactive and predictive decision-making rather than reactive HR intervention.
- To segment employees based on risk indicators such as job satisfaction, income levels, tenure, department, and performance metrics to identify high-risk attrition groups.
- To define and track strategic workforce KPIs including Attrition Rate, Retention Rate, Tenure Distribution, Compensation Trends, and Department-wise Attrition Impact.
- To apply predictive analytics using Azure ML to estimate employee attrition probability and identify high-risk employees in advance.
- To translate analytical and predictive findings into actionable retention insights that support workforce planning, talent optimization, and cost control.
- To develop an interactive Power BI decision-support dashboard that enables leadership to monitor retention health, compare segments, and evaluate potential intervention areas.

- To integrate workflow automation using Zapier for automated HR reporting and near real-time alert generation.
- To demonstrate how HR analytics can align with broader business strategy by linking employee retention insights to operational efficiency and long-term organizational growth.
- To document a structured end-to-end analytics workflow integrating data processing, prediction, visualization, and automation.

5. Methodology / Implementation Plan

#	Phase	Description
1	Data Collection & Exploration	The IBM HR Analytics dataset from Kaggle was analyzed to understand workforce structure and attrition patterns. Key variables such as department, job role, tenure, income, and satisfaction were reviewed to define the business problem and identify potential drivers of employee turnover.
2	Data Cleaning & Preprocessing	The dataset was cleaned to ensure accuracy and reliability for strategic analysis. Missing values were handled, categories standardized, and employee segments such as tenure bands and income groups were created for meaningful comparison.
3	Analysis & AI Integration	Strategic KPIs including Attrition Rate, Retention Rate, and Department-wise Attrition were defined using Power BI DAX. The analysis focused on identifying high-risk segments and patterns influencing attrition. Predictive modeling using Azure ML was applied to estimate attrition probability, while AI tools such as ChatGPT and Copilot were used to refine insights and strengthen interpretation.
4	Visualization / Output	An executive-level Power BI dashboard was developed to present attrition insights through KPIs, trend charts, and segmented views. The dashboard integrates analytical and predictive insights to support data-driven HR decision-making. Additionally, workflow automation using Zapier was implemented to trigger alerts based on high attrition risk, enabling near real-time reporting.
5	Documentation & Submission	The full workflow from data collection to analysis, prediction, and automated reporting was documented. Strategic findings and retention recommendations were compiled into a structured report with organized project files for submission.

6. Resources and Timeline

A. Hardware & Software Requirements

- Laptop — Intel(R) Core(TM) i3-7020U CPU (8 GB RAM)
- Operating System: Windows 11
- Microsoft Excel (Microsoft 365)
- Power BI Desktop — latest version
- Microsoft Copilot — via Microsoft 365 subscription
- AI Chatbot (ChatGPT / Google Gemini / Microsoft Copilot Web) — for prompt-based projects
- Internet connection — required for Copilot and AI Chatbot access
- Web browser — Google Chrome or Microsoft Edge

B. Dataset

- The project uses the **IBM HR Analytics Employee Attrition & Performance dataset** sourced from Kaggle, a publicly available HR analytics dataset widely used for workforce analysis.
- The dataset includes employee demographics, job roles, compensation details, performance indicators, tenure, job satisfaction, and attrition status.
- Data validation and preprocessing were conducted to ensure analytical reliability, including handling missing values, standardizing categorical fields, and verifying data consistency.
- Derived strategic metrics such as Attrition Rate, Tenure Bands, and Income Segmentation were created to enable workforce risk analysis.
- These transformations support identification of high-risk employee segments and enable the development of a data-driven retention framework through Power BI visualization.

C. Project Timeline

A simple table showing planned tasks and expected completion dates.

Phase	Task	Tentative Completion Date
1	Dataset collection, exploration, and understanding	<i>Week 1</i>
2	Data cleaning, preprocessing, and validation / Prompt design	<i>Week 2</i>

3	Analysis, AI tool integration (Copilot / Chatbot), and insight generation	<i>Week 2 – 3</i>
4	Dashboard / output build, testing, and review	<i>Week 3</i>
5	Report writing, GitHub upload, and video recording and submission	<i>Week 4</i>

7. Expected Outcomes / Conclusion

- A validated and analysis-ready HR dataset with documented data preparation processes to ensure reliability and scalability.
- Development of strategic workforce KPIs such as Attrition Rate, Retention Risk Segments, Average Tenure, and Compensation Distribution to support executive decision-making.
- An interactive Power BI dashboard designed as a decision-support tool, enabling leadership to monitor workforce health, identify high-risk segments, and evaluate attrition drivers.
- Implementation of predictive analytics using Azure ML to estimate employee attrition probability and enable early identification of high-risk employees.
- Clear identification of key factors influencing employee turnover, translated into actionable and predictive retention insights rather than descriptive analysis.
- Integration of workflow automation using Zapier to enable near real-time HR reporting and automated alert generation for high-risk employee segments.
- Application of AI tools such as Microsoft Copilot and ChatGPT to synthesize findings into concise business recommendations.
- A structured, data-driven retention framework that integrates analytics, prediction, and automation to support proactive decision-making and long-term workforce strategy.

8. References

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